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1.) Write general expressions for the electric and magnetic fields of a 1 GHz sinusoidal plane wave traveling in the +y direction in a lossless non-magnetic medium with relative permittivity $\epsilon_r = 9$. The electric field is polarized along the x direction, its peak value is 6 V/m, and its intensity is 4 V/m at t = 0 and y = 2 cm. (10 points)

2.) The electric field of a plane wave propagating in a lossless, nonmagnetic, dielectric material with $\epsilon_r = 2.56$ is given by

$$E = 20 \cos\left((6\pi \times 10^9)t - kz\right) \hat{y} \text{ (V/m)}. \qquad \text{(12 points total)}$$

Determine

(a) f, u_p, λ, k, and η. (5 points)

(b) the magnetic field H. (7 points)

3.) For a wave characterized by the electric field

$$\mathbf{E}(z,t) = a_x \cos(\omega t - kz)\hat{\mathbf{x}} + a_y \cos(\omega t - kz + \delta)\hat{\mathbf{y}},$$

identify the polarization state, determine the polarization angles (γ, χ), and sketch the locus of E(0, t) for each of the following cases. (16 points total)

(a) a_x = 3 V/m, a_y = 4 V/m, and δ = 0 radians (4 points)

(b) a_x = 3 V/m, a_y = 4 V/m, and δ = π radians (4 points)

(c) $a_x = 3$ V/m, $a_y = 3$ V/m, and $\delta = \pi/4$ radians (4 points)

(d) $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = -3\pi/4$ radians (4 points)

4. A linearly polarized plane wave of the form

$$\tilde{\mathbf{E}} = a_x e^{-jkz} \hat{\mathbf{x}}$$

can be expressed as the sum of an RHC-polarized wave with magnitude a_R and an LHC-polarized wave with magnitude a_L. Prove this statement by finding expressions for a_R and a_L in terms of a_x. (8 points)

5. Dry soil is characterized by $\epsilon_r = 2.5$, $\mu_r = 1$, and $\sigma = 10^{-4}$ S/m. At each of the following frequencies, determine if dry soil may be considered a good conductor, a quasi-conductor, or a low-loss dielectric. Then calculate α, β, λ, u_p, and η_c. (24 points total)

(a) 60 Hz (6 points)

(b) 1 kHz (6 points)

(c) 1 MHz (6 points)

(d) 1 GHz (6 points)

6. In a nonmagnetic, lossy, dielectric medium, a 300 MHz plane wave is characterized by the magnetic field phasor

$$\tilde{\mathbf{H}} = (\hat{\mathbf{x}} - j4\hat{\mathbf{z}})e^{-2y}e^{-j9y} \quad (\mathrm{A/m})$$

Obtain time-domain expressions for the electric and magnetic field vectors. (10 points)

7. A wave traveling in a nonmagnetic medium with ϵr = 9 is characterized by an electric field given by

$$\mathbf{E} = 3\cos\left(\left(\pi \times 10^7\right)t + kx\right)\hat{\mathbf{y}} - 2\cos\left(\left(\pi \times 10^7\right)t + kx\right)\hat{\mathbf{z}} \quad (\mathrm{V/m}).$$

Determine the direction of wave travel and average power density carried by the wave. (10 points)